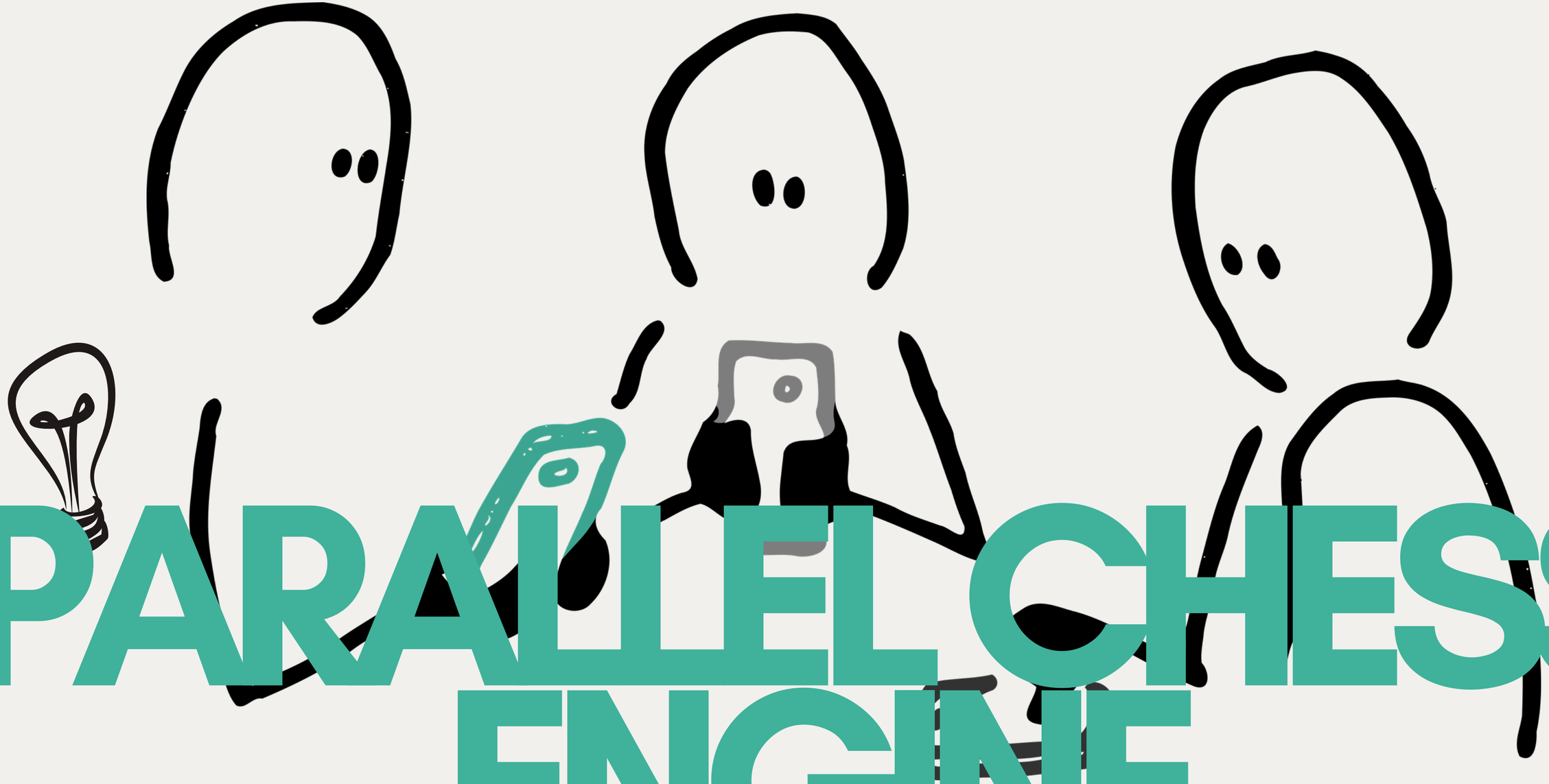


PARALLEL CHESS ENGINE

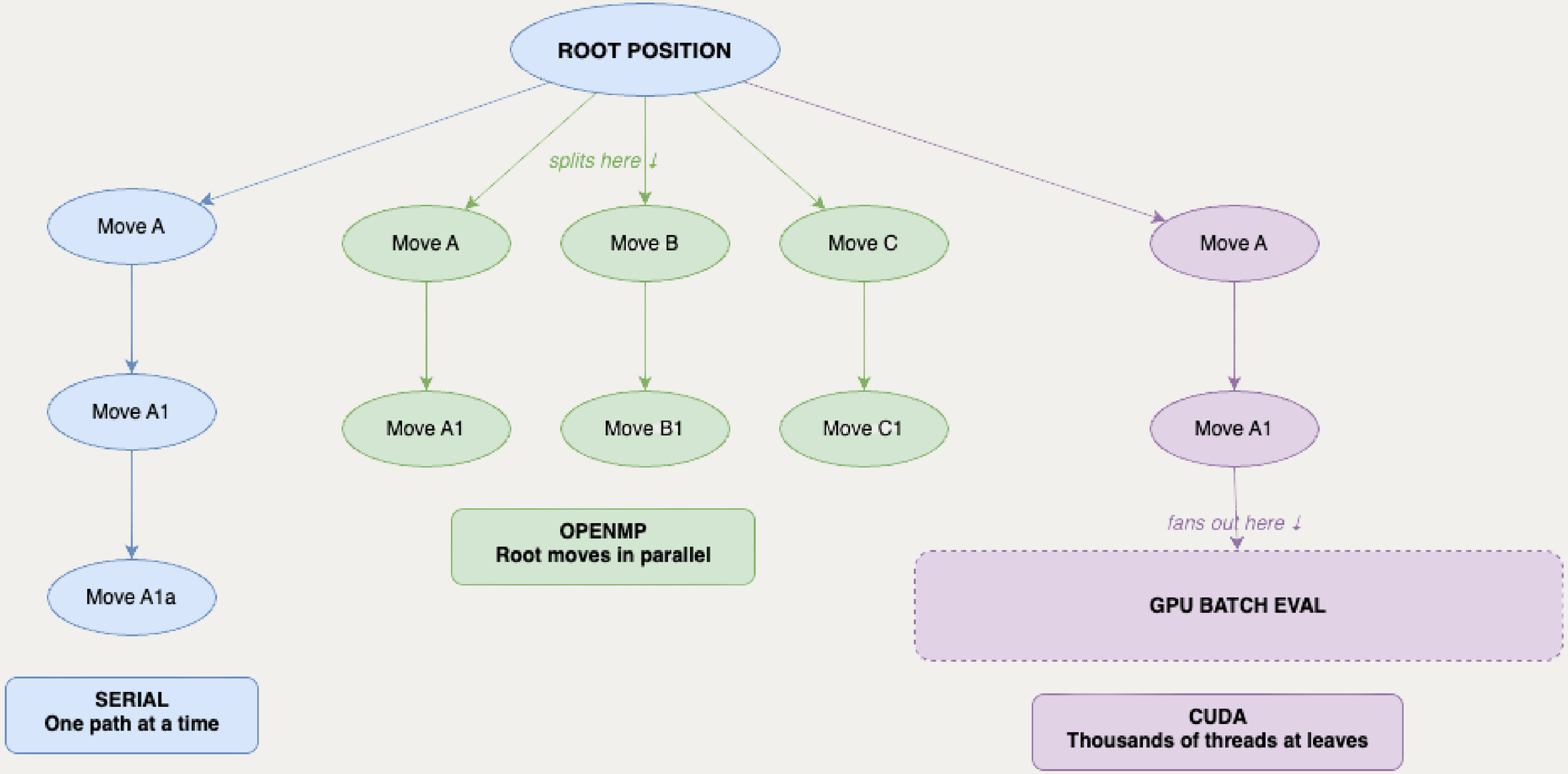
GURNOOR SINGH PANNU
RAMYAK SHARMA
KRISHNA JINDAL

WHY IS YOUR CHESS ENGINE SO SLOW BRO?

At depth 8, a chess engine evaluates 19 million positions. Serial CPU does this one by one. That's not computing. That's suffering.

WHAT WE BUILT:

THREE ENGINES. ONE BOARD.



OPENMP: PARALLEL AT THE ROOT

Depth	Serial	OpenMP	SpeedUp
3	6.5	5.5	1.17×
4	37.9	34.5	1.10×
5	194.3	172.9	1.12×
6	791.5	713.2	1.11×
7	2835.1	2857.1	0.99×
8	21650.3	19689.2	1.10 x

Consistent ~1.1× speedup across depths

Depth 7 anomaly: thread scheduling overhead briefly exceeded gains — expected on 2-core Colab

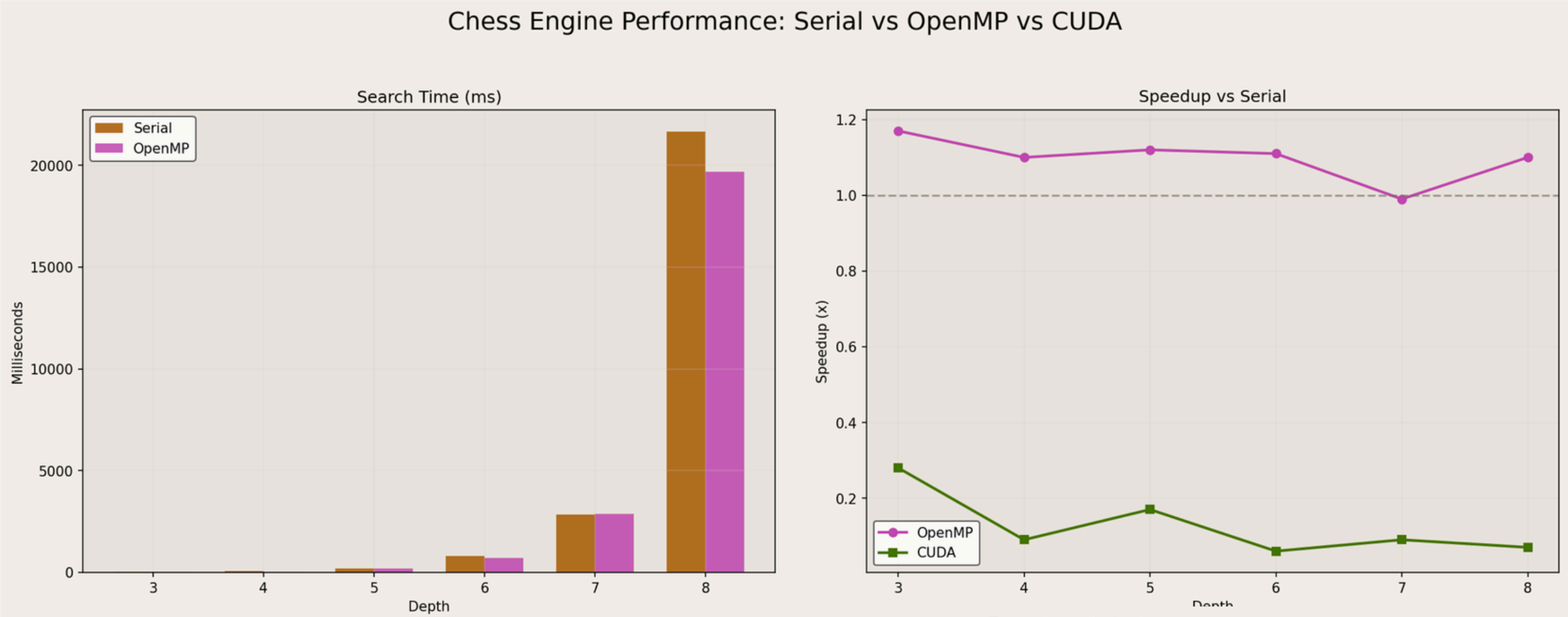
CUDA RESULTS:

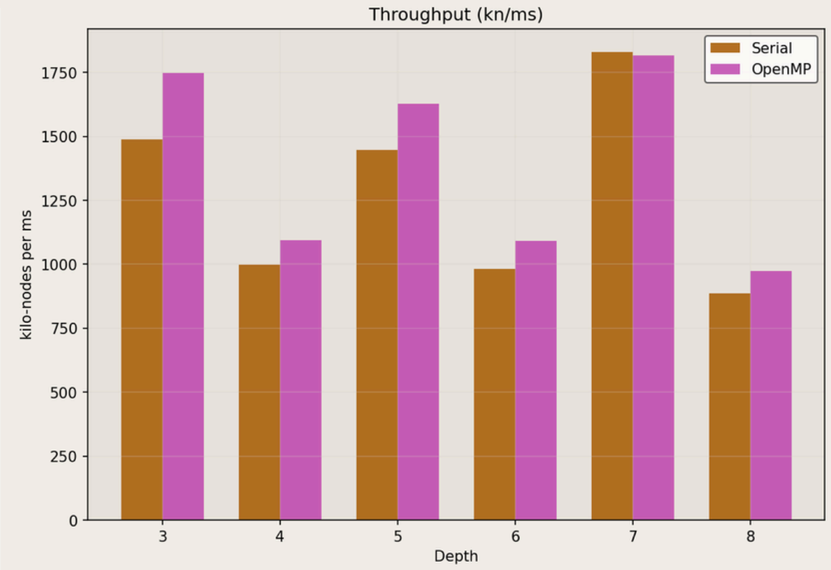
Depth	Serial	CUDA	SerialNodes	CUDA Nodes
5	194	1,125	281,539	279,340
6	791	12186	778,569	2,921,085
7	2835	30,818	5,189,985	6,684,720
8	21,650	299,320	19,198,976	75,133,003

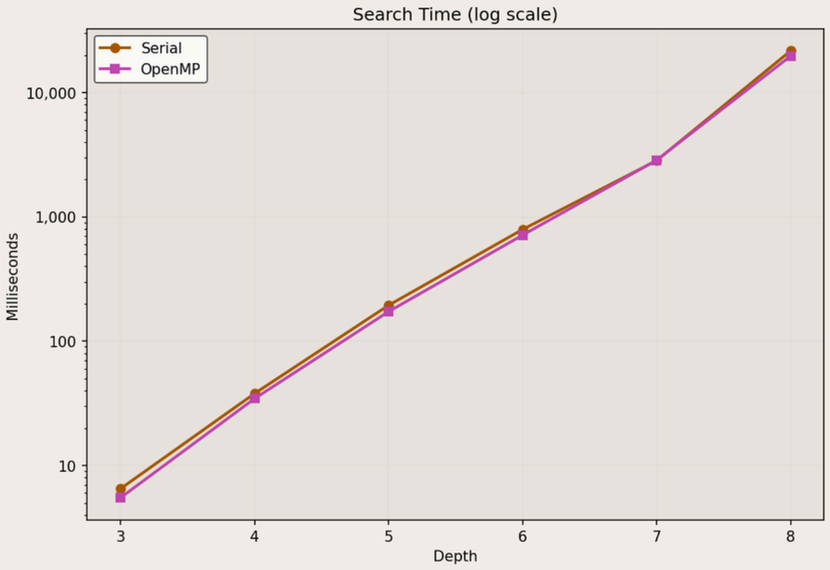
At depth 8, CUDA explored 75 million nodes vs Serial's 19 million — 4× more ground covered

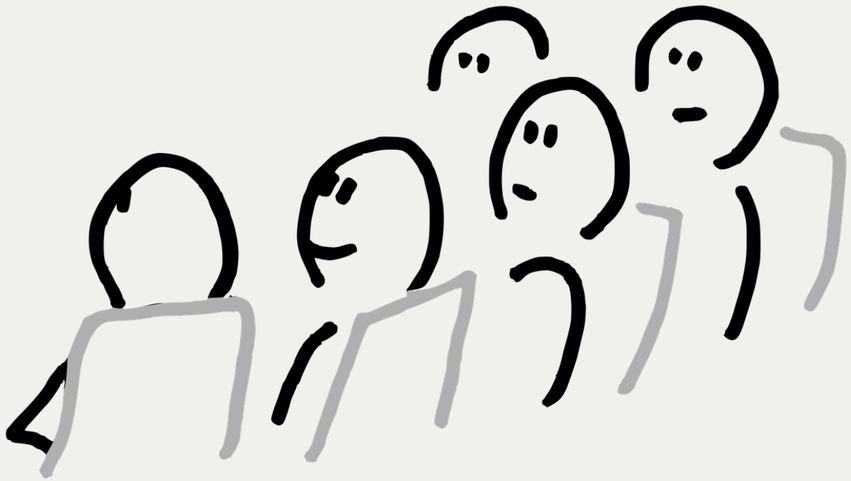
Faster exploration. Slower execution.

GRAPHICAL REPRESENTATION:







OPENMP TO CUDA :

THANK YOU!